Linear Programming Approach to the Graph Square Problem

Edward Xu

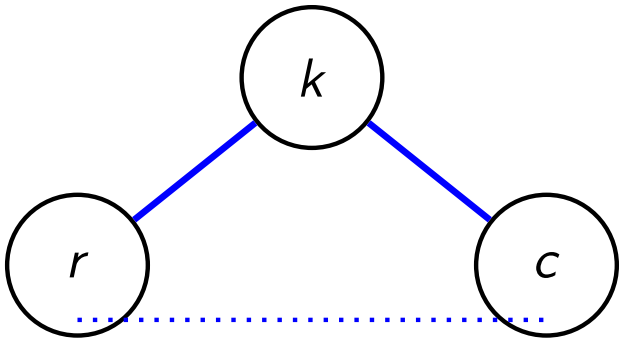
University of Melbourne

Supervisor: Dr Christopher Duffy

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Square of a Graph

Two-step connections

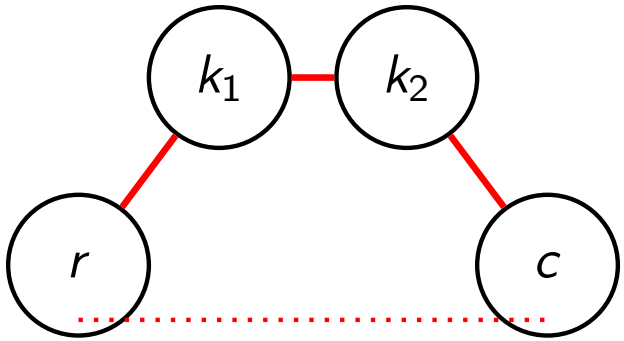


two steps

$$(A^2)_{rc} = \sum_{k=1}^{n} A_{rk} A_{kc}.$$

Counts walks of length two from r to c.

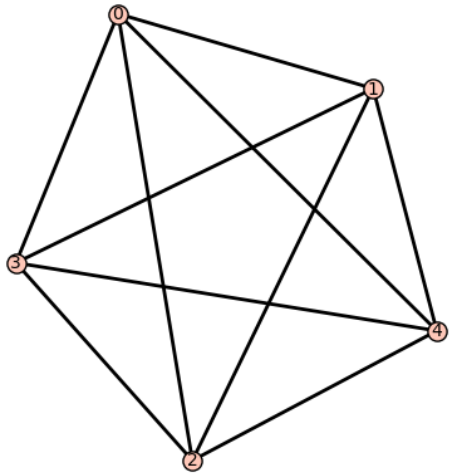
Three-step connections



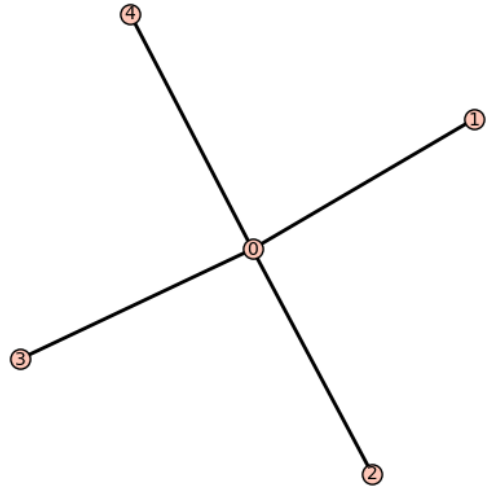
three steps

$$(A^3)_{rc} = \sum_{k=1}^{n} A_{rk} (A^2)_{kc}.$$

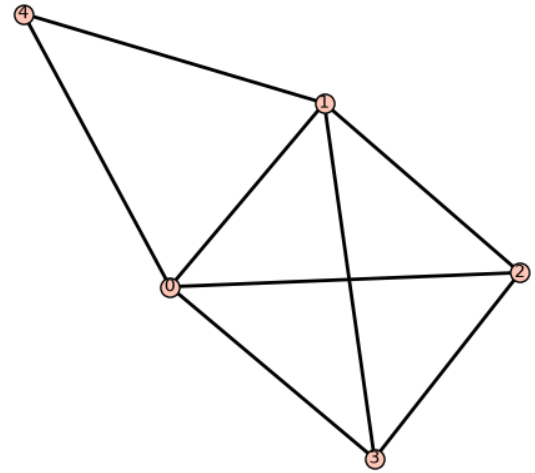
Counts walks of length three from r to c.



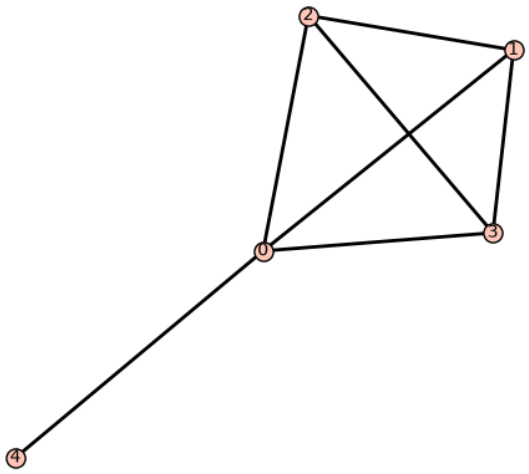
(a) K_5, which is a root of itself.



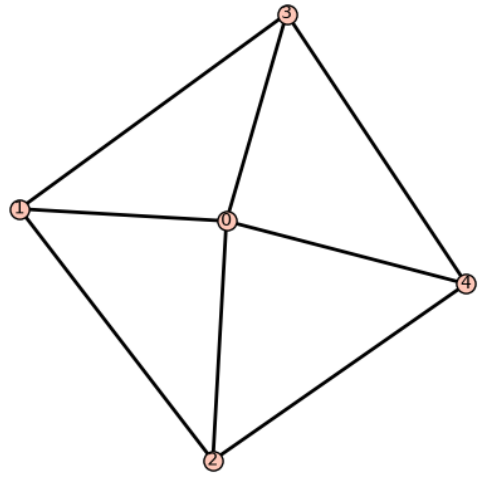
(b) $K_{1,4}$, a root of K_5.



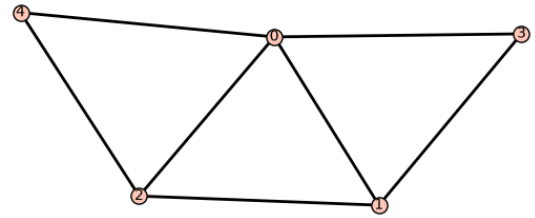
(c) A root of K_5.



(d) A root of K_5.



(e) A root of K_6.



(f) A root of K_7.

Figure: A few of the 15 non-isomorphic graph roots of K_5.

The Graph Square Problem

Problem (SQUARE)

Instance: *A finite, simple, undirected graph* $G = (V, E)$.

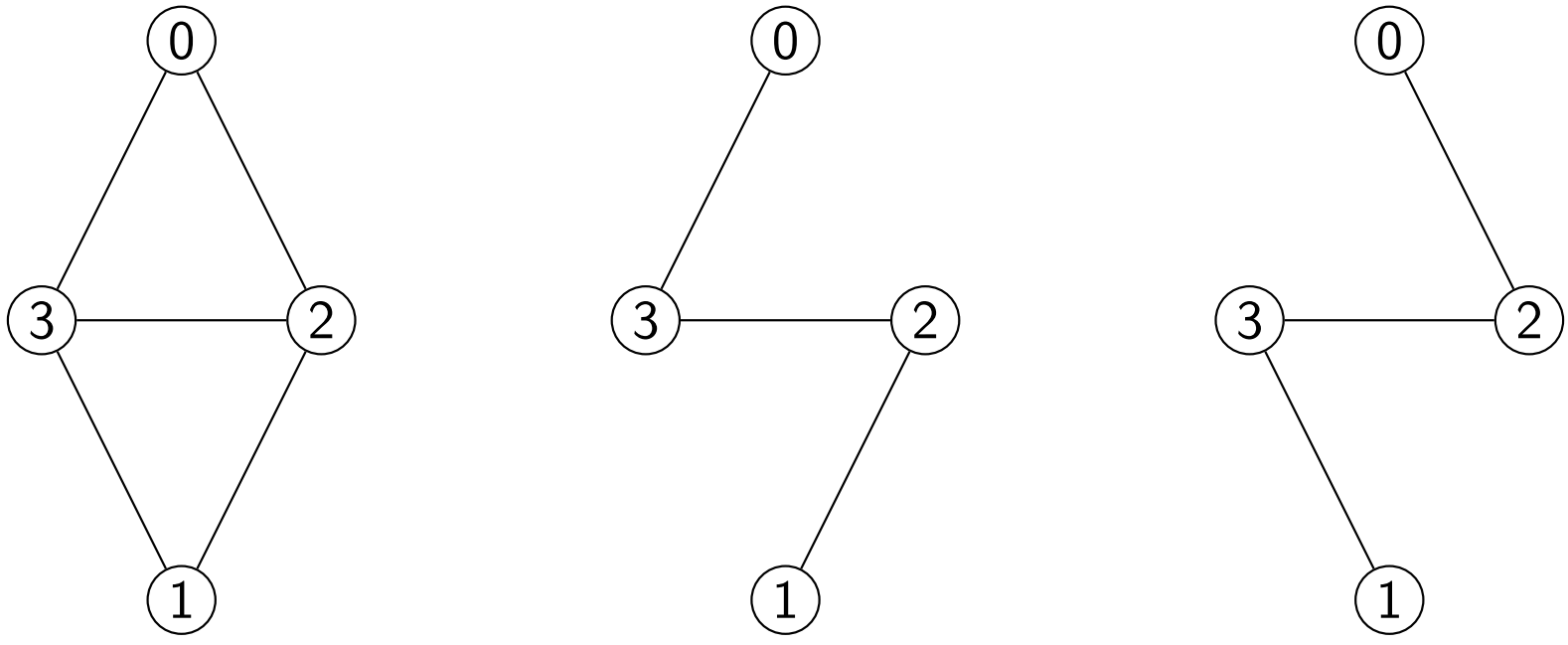
Question: *Does there exist a finite, simple, undirected graph* H *such that*

$$G = H^2?$$

Let the **graph root** $H = (V, E_H)$ be a finite, simple, undirected graph, where V is the vertex set with $|V| = n$, and $E_H \subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}$ is the edge set. The **square of a graph** H, denoted by $H^2 = (V, E_{H^2})$ is the graph on the same vertex set with edge set

$$E_{H^2} = \{uv \mid u \neq v, 1 \leq d_H(u, v) \leq 2\}. \tag{1}$$

When $G = H^2$ we say that H is a **root** of G.



(a) G (b) T_1 (c) T_2

Figure: The graph G has two possible P_4 square roots that are trees, T_1 and T_2. The two roots are isomorphic, differing only by the interchange of vertices 2 and 3.

NP-Completeness

Motwani and Sudan (1994) proved that the graph square problem is **NP**-complete by showing that the problem is **NP**-hard through a polynomial reduction from Not-All-Equal 3-SAT.

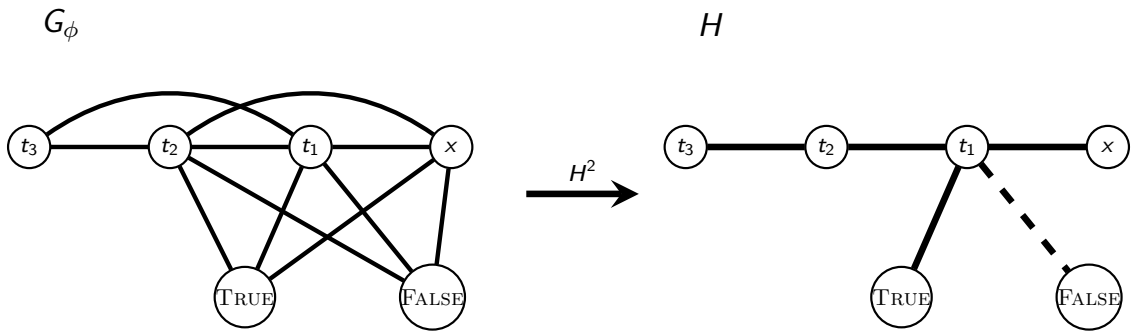
Problem (NAESAT)

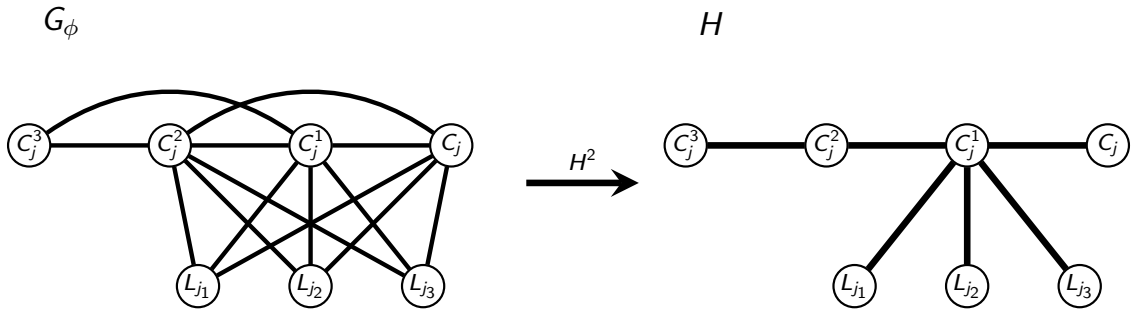
Instance: *A 3-CNF formula*

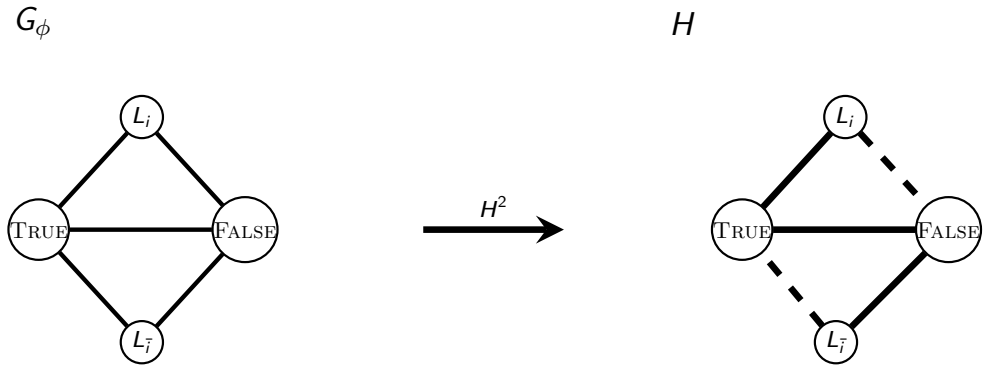
$$\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_m$$

where each clause C_i is over variables $x_1, \ldots, x_n$.

Question: *Does there exist an assignment of* TRUE *and* FALSE *to the variables such that every clause contains at least one* TRUE *literal and at least one* FALSE *literal?*


G_ϕ
H
t_3
t_2
t_1
x
True
False
H^2


G_ϕ
H
C_j^3
C_j^2
C_j^1
C_j
L_{j_1}
L_{j_2}
L_{j_3}
H^2


G_ϕ
H
L_i
True
False
$L_{\bar{i}}$
H^2

Restricting the Square vs the Root

Square G	**Root H**	**Main result**
Unrestricted	Trees	Ross and Harary (1960): structural characterization and reconstruction of tree roots, unique up to isomorphism [RH60].
Unrestricted	Trees	Lin and Skiena (1995): optimal linear-time recognition and reconstruction of tree roots [LS95].
2-connected	Unrestricted	Fleischner (1974): the square of a 2-connected graph has a Hamiltonian cycle [Fle74].
Interval squares	Unrestricted	Lau and Corneil (2004): polynomial-time construction of roots from interval squares [LC04].
Unrestricted	Block graphs	Le and Tuy (2007): linear-time recognition of roots of block graphs [LT07].
Unrestricted	Bipartite	Lau (2006): linear-time construction of bipartite roots [Lau06].
Directed graphs	Directed graphs	Geller (1970): necessary and sufficient conditions for directed square-root existence [Gel70].
Graphs of prescribed girth	Unrestricted	Farzad, Lau, Le, and Tuy (2009) and Farzad and Karimi (2012): complexity dichotomy for square-root recognition [Far+09; FK12].
Maximum degree ≤ 6	Unrestricted	Cochefert, Couturier, Golovach, Kratsch, and Paulusma (2013): minimum-edge roots in polynomial time and maximal-edge roots in exponential time [Coc+13].
Unrestricted	Oriented graphs, maximum degree ≤ 3, girth ≥ 4	Duffy (2025): classification of squares of oriented graphs under these root restrictions [Duf25].

Maximal Cliques and Unicliqual Vertices in Tree Roots

Input: A graph G where $G = T^2$
Output: A tree T
Step 1
if $G = K_p$ is complete **then**
    Return tree T with depth 1, root any vertex
else
    Continue
end if
Step 2
Identify all maximal cliques $C_1, \ldots, C_n$ of G (e.g. Bron–Kerbosch [BK73])
Step 3
Let $b_1, \ldots, b_n$ be multicliqual points for $C_1, \ldots, C_n$
Build graph T on $\{b_1, \ldots, b_n\}$
for each pair (C_i, C_j) with $|C_i \cap C_j| = 2$ **do**
    Add edge (b_i, b_j) in T
end for
T is a tree
Step 4
for each clique C_i **do**
    Let n_i be the number of unicliqual points in C_i
    Attach n_i leaves to b_i in T
end for
Resulting T has $|V(G)|$ nodes and $T^2 = G$

Integer Program

Two distinct vertices are adjacent in $G = H^2$ if and only if they are connected by a path of length one or two in H.

$$A_G = (A_H)^2 \vee A_H. \tag{2}$$

$$(A_H)^2_{ij} = \big((A_H)_{i1} \wedge (A_H)_{1j}\big) \vee \big((A_H)_{i2} \wedge (A_H)_{2j}\big) \vee \cdots \vee \big((A_H)_{in} \wedge (A_H)_{nj}\big)$$

$$g_{ij} \text{ if and only if } \Big(\bigwedge_k (x_{ik} \wedge x_{kj}) \vee x_{ij}\Big). \tag{3}$$

Lemma

Let G and H be graphs with adjacency matrices A_G and A_H,respectively. The graph H is a root of G if and only if

$$A_G = \mathbb{1}_{\geq 1}\big((A_H)^2 - D_H + A_H\big).$$

i) $x_{ij} \in \{0,1\}$ denotes the (i,j) entry of the adjacency matrix A_H.

ii) $y_{ikj} \in \{0,1\}$ represents the product $x_{ik}x_{kj}$ in $(A_H)^2_{ij}$.

iii) $z_{ij} \in \mathbb{Z}_{\geq 0}$ represents the argument of the indicator function.

iv) $M \in \mathbb{Z}_{>0}$ be a sufficiently large constant $M \leq n(n-1) = O(n^2)$.

$$D_H = \begin{bmatrix} \deg_H(1) & 0 & \cdots & 0 & 0 \\ 0 & \ddots & \vdots & \ddots & \vdots \\ \vdots & \cdots & \deg_H(i) & \cdots & 0 \\ 0 & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & \deg_H(n) \end{bmatrix}, \quad A_H = \begin{bmatrix} x_{11} & \cdots & x_{1j} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i1} & \cdots & x_{ij} & \cdots & x_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nj} & \cdots & x_{nn} \end{bmatrix}$$

$$(A_H)^2 = \begin{bmatrix} \sum_k x_{1k}x_{k1} & \cdots & \sum_k x_{1k}x_{kj} & \cdots & \sum_k x_{1k}x_{kn} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \sum_k x_{ik}x_{k1} & \cdots & \sum_k x_{ik}x_{kj} & \cdots & \sum_k x_{ik}x_{kn} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \sum_k x_{nk}x_{k1} & \cdots & \sum_k x_{nk}x_{kj} & \cdots & \sum_k x_{nk}x_{kn} \end{bmatrix}.$$

$$\begin{bmatrix} g_{11} \\ \vdots \\ g_{1j} \\ \vdots \\ g_{1n} \\ \vdots \\ g_{ij} \\ \vdots \\ g_{nn} \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{\geq 1}(z_{11}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{1j}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{1n}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{ij}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{nn}) \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{1k}x_{k1} - (D_H)_{11} + x_{11}\big) \\ \vdots \\ \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{1k}x_{kj} - (D_H)_{1j} + x_{1j}\big) \\ \vdots \\ \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{1k}x_{kn} - (D_H)_{1n} + x_{1n}\big) \\ \vdots \\ \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{ik}x_{kj} - (D_H)_{ij} + x_{ij}\big) \\ \vdots \\ \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{nk}x_{kn} - (D_H)_{nn} + x_{nn}\big) \end{bmatrix}. \tag{4}$$

Objective Function:

$$\max \quad 0 \tag{5}$$

Variables:

$$x_{ij} \in \mathbb{Z}_{\geq 0}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{6}$$

$$y_{ikj} \in \mathbb{Z}_{\geq 0}, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{7}$$

$$z_{ij} \in \mathbb{Z}_{\geq 0}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{8}$$

Constraints:

(1) Indicator argument:

$$z_{ii} = \sum_{k=1}^{n} y_{iki} - \sum_{k=1}^{n} x_{ik}, \forall i \in \{1, \ldots, n\}, \tag{9}$$

$$z_{ij} = \sum_{k=1}^{n} y_{ikj} + x_{ij}, \qquad \forall i \neq j \in \{1, \ldots, n\}, \tag{10}$$

(2) Indicator function:

$$z_{ij} \leq M g_{ij}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{11}$$

$$z_{ij} \geq g_{ij}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{12}$$

(3) Product:

$$y_{ikj} \leq x_{ik}, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{13}$$

$$y_{ikj} \leq x_{kj}, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{14}$$

$$y_{ikj} \geq x_{ik} + x_{kj} - 1, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{15}$$

(4) Upper Bound:

$$x_{ij} \leq 1, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{16}$$

$$y_{ikj} \leq 1, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{17}$$

(5) Symmetry:

$$x_{ij} = x_{ji}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{18}$$

(6) Diagonal:

$$x_{ii} = 0, \qquad \forall i \in \{1, \ldots, n\}. \tag{19}$$

The integer program on the arbitrary graph G using Equations (6)–(19) together with the constant objective function Equation (5) is denoted as $\mathcal{IP}(G)$.
The total number of simple graphs on $n = 2, 3, 4, \ldots$ unlabeled nodes is the sequence $2, 4, 11, 34, 156, 1044, 12346, \ldots$ given by A000088. The number of graphs which are squares is the sequence $2, 3, 6, 11, 28, 77, 307, \ldots$ given by A382181.

n	H exists	H does not exist
3	3	1
4	6	5
5	11	23
6	28	128
7	77	967
8	307	12039

Table: Outcomes of the integer program for graphs of different orders of n.

Methodology

Implemented in **SageMath**, with **Gurobi** as the optimization backend.

Uses the **barrier method**, an interior-point method based on predictor–corrector Newton iterations.

Graphs are considered up to isomorphism, but algorithms operate on labelled representatives generated by SageMath.

Vertex labels are retained throughout each computation.

Different labellings of the same graph may lead to different intermediate or selected optimization solutions.

Results correspond to the labelled representatives produced by the specified generation procedure.

Code and test harness:
`https://github.com/Hellwoodoo/Graph-Squares`.

Linear Program

Listing: Linear Program.

```
def linear_program(G):
    m = MixedIntegerLinearProgram(solver='Gurobi', maximization=False)
    m.solver_parameter("Method", 2)
    x = m.new_variable(real=True, nonnegative=True)
    y = m.new_variable(real=True, nonnegative=True)
    z = m.new_variable(real=True, nonnegative=True)
    A_G = G.adjacency_matrix()
    n = A_G.nrows()
    M = n**2
    for i in range(n):
        for j in range(n):
            if i == j:
                m.add_constraint(z[i,j] - (sum(y[i,k,j] for k in range(n)) - sum(x[i,k] for k in range(n)) + x[
   i,j]) == 0)
            else:
                m.add_constraint(z[i,j] - (sum(y[i,k,j] for k in range(n)) + x[i,j]) == 0)
    for i in range(n):
        for j in range(n):
            m.add_constraint(z[i,j] - M * A_G[i][j] <= 0)
            m.add_constraint(-z[i,j] + A_G[i][j] <= 0)
    for i in range(n):
        for j in range(n):
            for k in range(n):
                m.add_constraint(y[i,k,j] - x[i,k] <= 0)
                m.add_constraint(y[i,k,j] - x[k,j] <= 0)
                m.add_constraint(- y[i,k,j] + x[i,k] + x[k,j] - 1 <= 0)
    for i in range(n):
        for j in range(n):
            m.add_constraint(x[i,j] == x[j,i])
    for i in range(n):
        m.add_constraint(x[i,i] == 0)
    for i in range(n):
        for j in range(n):
            m.add_constraint(x[i,j] - 1 <= 0)
            for k in range(n):
                m.add_constraint(y[i,k,j] - 1 <= 0)
    return m, x
```

Linear Relaxation

$$x_{ij} \in \mathbb{R}_{\geq 0}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{20}$$

$$y_{ikj} \in \mathbb{R}_{\geq 0}, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{21}$$

$$z_{ij} \in \mathbb{R}_{\geq 0}, \qquad \forall i, j \in \{1, \ldots, n\}. \tag{22}$$

Such a linear program may have *integral solutions* and *fractional solutions* when the root exists and *fractional solutions* and *infeasible solutions* when the root does not exist.

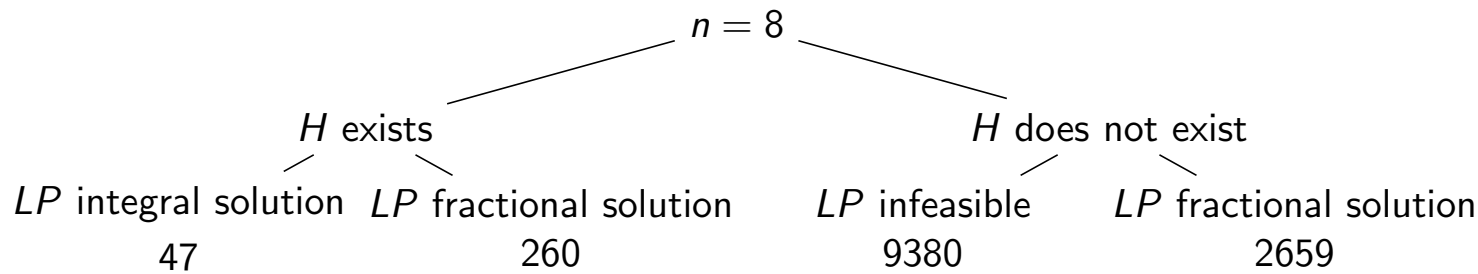
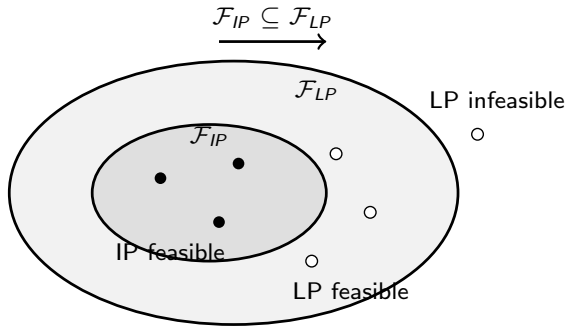
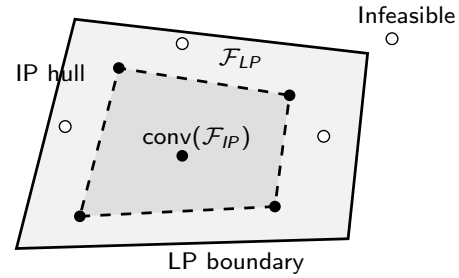


Figure: LP outcomes for $n = 8$ vertices using $\mathcal{LP}(G)$.



Set-theoretic



Geometric

The LP integral solution sequence $2, 3, 6, 10, 17, 29, 51, \ldots$ matches A285665. However, it is unclear why this should be the case.

n	H exists		H does not exist	
	LP integral solution	LP fractional solution	LP infeasible	LP fractional solution
3	3	0	1	0
4	5	1	5	0
5	7	4	20	3
6	13	15	107	21
7	22	55	767	200
8	44	263	9380	2659

Table: Outcomes of the LP relaxation for graphs of different orders n, classified according to whether a square root H exists.

The linear relaxation will never return an infeasible solution.

Proposition

If G has a square root H, then $\mathcal{LP}(G)$ is feasible.

Proof.

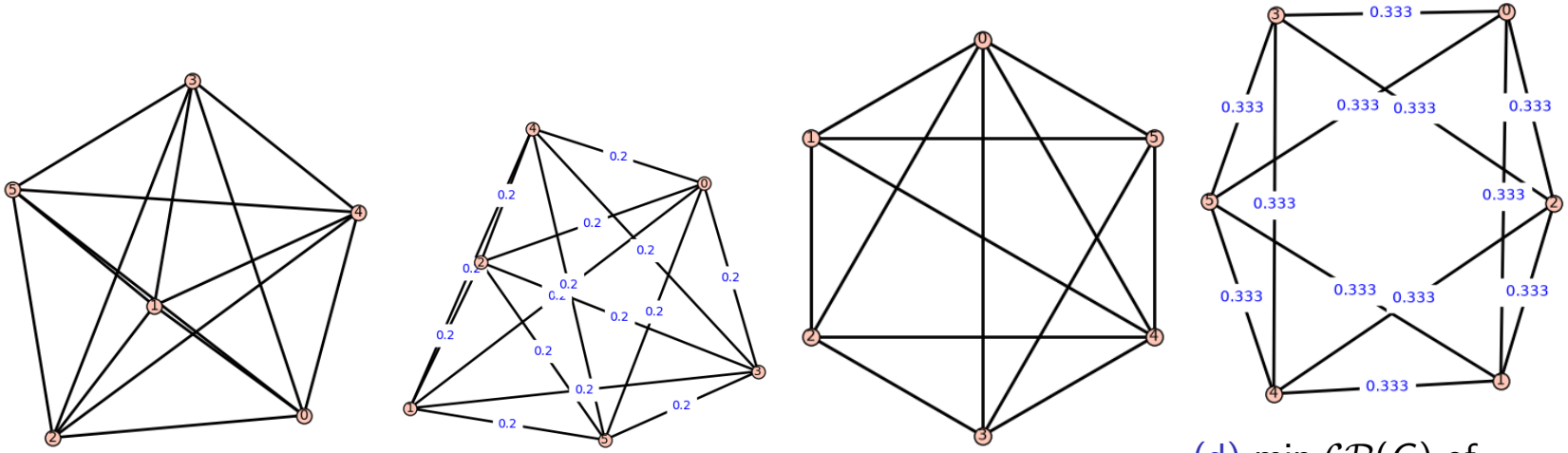
Since $\mathcal{LP}(G)$ is obtained by relaxing the integrality constraints of $\mathcal{IP}(G)$, every feasible solution of $\mathcal{IP}(G)$ is also feasible for $\mathcal{LP}(G)$. Hence $\mathcal{LP}(G)$ is feasible. □

Min and Max Objective Function

$$\min \sum_{i=1}^{n} \sum_{j=1}^{n} x_{ij}. \tag{23}$$

n	H exists		H does not exist	
	LP integral solution	LP fractional solution	LP infeasible	LP fractional solution
3	2	1	1	0
4	3	3	5	0
5	3	8	20	3
6	4	24	107	21
7	5	72	767	200
8	7	300	9380	2659

Table: LP outcomes for different values of n using the linear program with a minimisation objective function.



(a) K_6. (b) $\min \mathcal{LP}(G)$ of K_6. (c) $K_6 \backslash \{13, 25\}$. (d) $\min \mathcal{LP}(G)$ of $K_6 \backslash \{13, 25\}$.

Figure: For K_6, the solution to $\min \mathcal{LP}(G)$ is completely fractional, with every edge having weight $\frac{1}{5}$. For $K_6 \backslash \{13, 25\}$, the solution is completely fractional, with every edge having weight $\frac{1}{3}$.

Conjecture (Fractional Cliques in $\min \mathcal{LP}(G)$)

Let G be a graph with $n \geq 3$ vertices. For every edge $uv \in E_G$, let t_{uv} be the number of vertices in a smallest maximal clique of G containing uv. Then $\min \mathcal{LP}(G)$ *returns the solution given by*

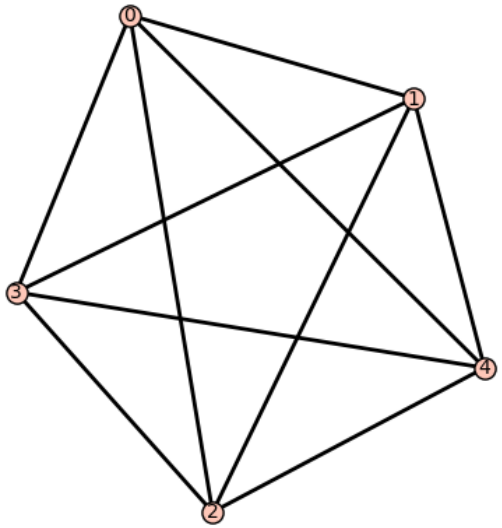
$$x_{uv} = \frac{1}{t_{uv} - 1} \quad \text{for all } uv \in E_G.$$

$$\max \sum_{i=1}^{n} \sum_{j=1}^{n} x_{ij}. \tag{24}$$

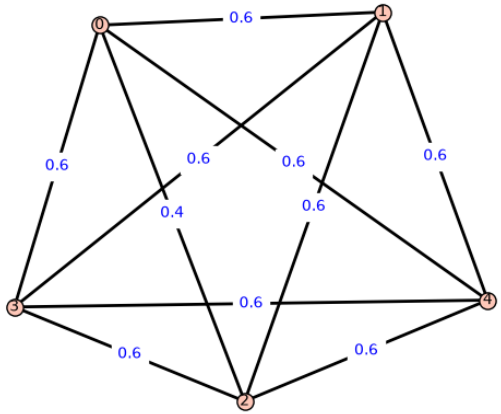
n	H exists		H does not exist	
	LP integral solution	LP fractional solution	LP infeasible	LP fractional solution
3	3	0	1	0
4	6	0	5	0
5	9	2	20	3
6	18	10	107	21
7	38	39	767	200
8	94	213	9380	2659

Table: LP outcomes for different values of n using the linear program with a maximisation objective function.

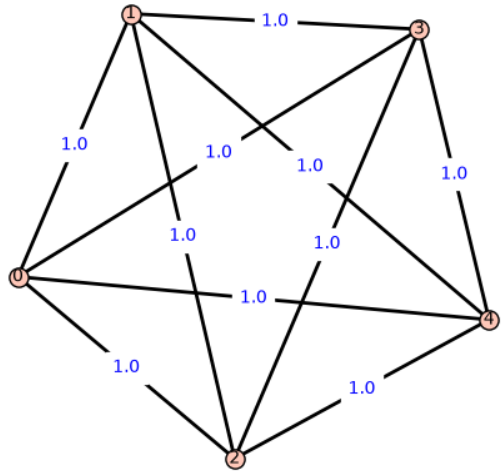
The single complete graph K_3 is selected as the unique optimal solution.



(a) K_5.



(b) Linear program result for K_5.



(c) Max linear program result for K_5.

Figure: The graph K_3 and two of its square roots.

Fundamental Properties

Lemma

Let G and H be graphs such that $G = H^2$. Let $a, b \in V_G$ such that $ab \notin E_G$. For every feasible solution x of $\mathcal{LP}(G)$ and every $v \in V_G$ with $v \neq a, b$ we have $x_{ab} = y_{avb} = 0$.

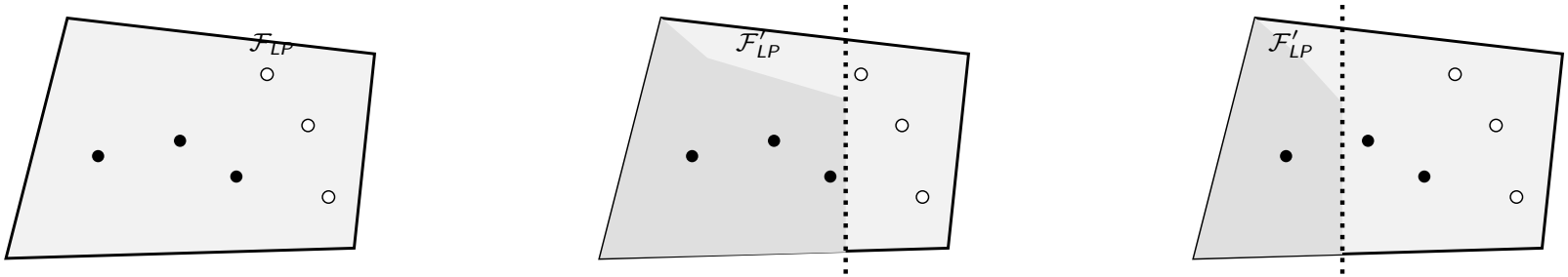
Lemma

Let G and H be graphs such that $G = H^2$. Let $a, b \in V_G$ such that $ab \notin E_G$. For every feasible solution x of $\mathcal{LP}(G)$ and every $v \in V_G$ with $v \neq a, b$ we have

$$x_{av} + x_{vb} \leq 1.$$

Lemma

Let G and H be graphs such that $G = H^2$. Let $a, b, v \in V_G$ such that $ab \notin E_G$. If a feasible solution x of $\mathcal{LP}(G)$ satisfies $x_{av} = 1$, then $x_{vb} = 0$.



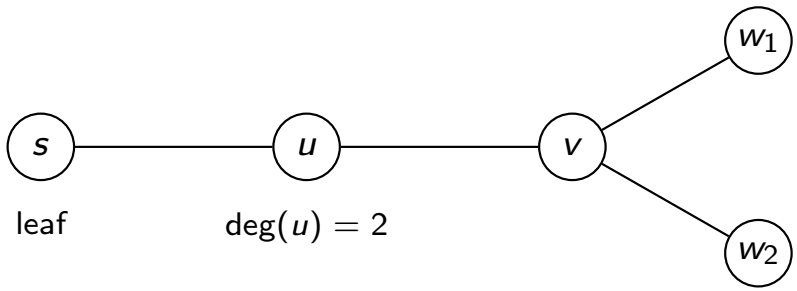
Base LP **Useful strengthening** **Incorrect strengthening**

A suitable constraint eliminates fractional solutions while retaining all valid integer solutions, whereas an overly restrictive constraint also excludes valid integer solutions.

Identifying Square Characteristics of Tree Roots

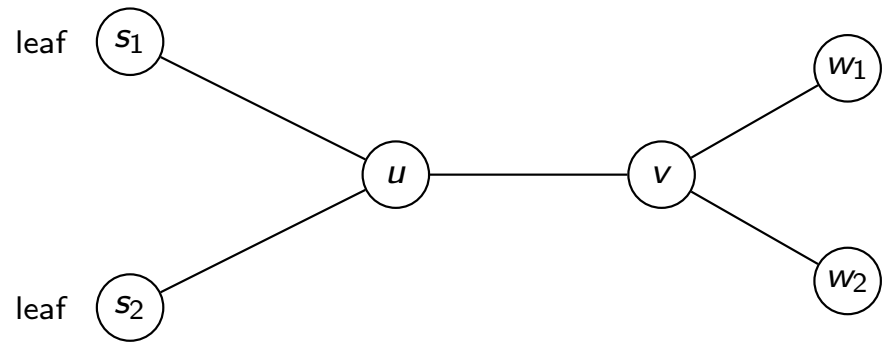
Definition (Sparse Pendant)

Let H be a graph. A **sparse pendant** is a path $s - u - v$ in H such that s is a leaf and u has degree 2.



Definition (Pendant Fork)

Let H be a graph. A **pendant fork** is the induced subgraph of H on four vertices s_1, s_2, u, v, where s_1 and s_2 are leaves adjacent to u, and every neighbour of u other than v is a leaf.



Sparse Pendants or Pendant Forks in Trees

Lemma

Every non-trivial tree with at least 3 vertices contains either a sparse pendant or a pendant fork subgraph.

Proof.

Root T at any vertex. Let s be a leaf of maximum depth, u its parent. Every child of u is a leaf, else a deeper vertex than s would exist. **Case 1: u is not the root.** Let v be the parent of u. Every neighbour of u other than v is a leaf child.

If u has exactly one leaf neighbour s, then $\deg_T(u) = 2$, and $\{s, u, v\}$ induces a sparse pendant.

If u has two leaf neighbours s_1, s_2, then $\{s_1, s_2, u, v\}$ induces a pendant fork.

Case 2: u is the root. All neighbours of u are leaf children, and u has at least two of them.

If u has exactly two neighbours s, v, then $\{s, u, v\}$ induces a sparse pendant.

Otherwise, pick distinct children s_1, s_2, v of u; then $\{s_1, s_2, u, v\}$ induces a pendant fork.

Leaf Fixing Linear Program

If s is a leaf, u is its corresponding stem, and there exists $v \in N_G[u] \setminus \{u\}$ such that

$$N_G[s] \subset N_G[u] \subseteq N_G[v],$$

with all vertices in $N_G[s] \setminus \{u, v\}$ having the same closed neighbourhood as s, then the corresponding root edges can be fixed by imposing

$$x_{su} = x_{us} = x_{uv} = x_{vu} = 1. \tag{25}$$

Sparse pendant: s belongs to a unique K_3 in G.

Pendant fork: An induced $K_4 = \{s_1, s_2, u, v\}$, where s_1, s_2 are leaves of u, satisfies

$$N_G(s_1) \setminus K_4 = N_G(s_2) \setminus K_4 \subset N_G(u) \setminus K_4 \subseteq N_G(v) \setminus K_4.$$

Lemma

Let G and T be graphs such that $G = T^2$ and T is a tree. $\max \mathcal{UVLP}(G)$ always fixes a sparse pendant or pendant fork subgraph of T.

Whenever a root H of G exists, the addition of the constraint Equation (25) preserves the feasibility of the linear program.

Proposition (Validity of Unicliqual Vertex Fix)

If a root H of G exists, then $\max \mathcal{UVLP}(G)$ *is feasible.*

n	H exists		H does not exist	
	LP integral solution	LP fractional solution	LP infeasible	LP fractional solution
3	3	0	1	0
4	6	0	5	0
5	10	1	21	2
6	22	6	117	11
7	51	26	852	115
8	148	159	10271	1768

Table: Outcomes of the unicliqual vertex linear program for graphs of different orders n.

Propagation of the Fixing

Lemma (0 Path Propagation)

Let G and H be graphs such that $G = H^2$. Let $a, i, j, b \in V_G$ such that a, i, j, b form a shortest induced path in H between a and b. For every feasible solution x of $\mathcal{LP}(G)$,

$$x_{ai} = 1 \implies x_{ib} = 0, \quad x_{ia} = 1 \implies x_{bi} = 0, \quad x_{jb} = 1 \implies x_{aj} = 0, \quad x_{bj} = 1$$

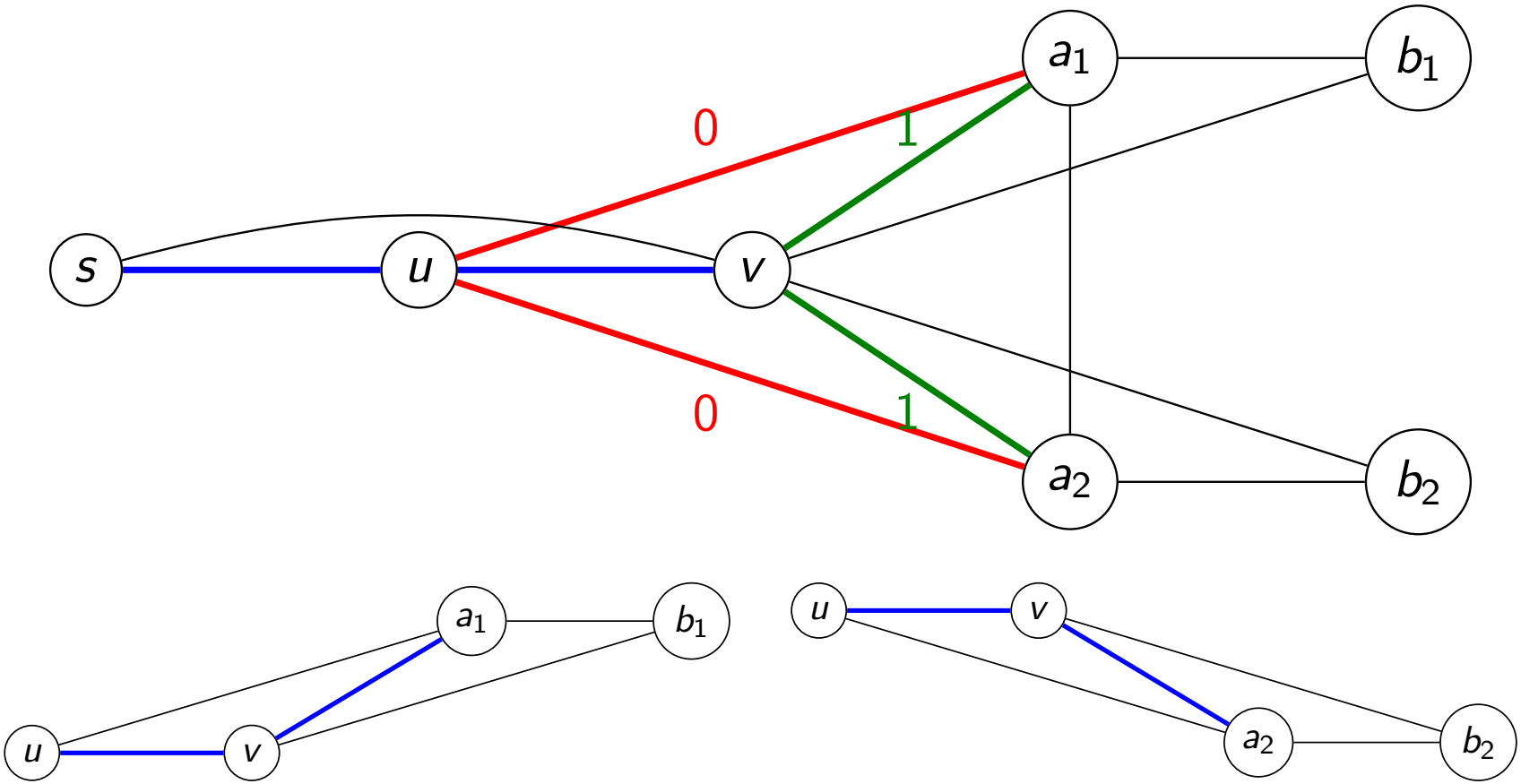
Lemma (1-Path Propagation)

Let G be a graph and let x be a feasible solution of $\mathcal{LP}(G)$. Suppose that $u - v - a_1$ form a shortest induced path in H between u and a and that $x_{uv} = 1$. Let $A = \{a_1, \ldots, a_r\}$ and $B = \{b_1, \ldots, b_s\}$ be such that every vertex $k \in V_G \setminus \{u, v\}$ satisfies $k \in A \cup B$, and suppose that

$$x_{um} = 0 \text{ for every } m \in A, \quad g_{un} = 0 \text{ for every } n \in B,$$

and $g_{sa_1} = 0$ for the vertex s preceding u on the shortest path to a_1. Then

$$x_{va_1} = 1.$$


a_1
b_1
0
1
s
u
v
0
1
a_2
b_2
a_1
b_1
u
v
u
v
a_2
b_2

Vertex Deletion

Input: A graph G
Output: A LP model (m, x)
Construct and solve $\mathcal{LP}(G)$.
if $\mathcal{LP}(G)$ is integral **then**
    return (m, x)
end if
Set $F = \left\{ \{a, b\} \in \binom{V_G}{2} : x_{ab} = x_{ba} = 1 \right\}$.
Compute $d_F(u) = |\{w \in N_G(u) : \{u, w\} \notin F\}|$ for each $u \in V_G$.
Choose $v \in \arg\min_{u \in V_G} d_F(u)$ and set $V^- = V_G \setminus \{v\}$.
Construct $G^- = G[V^-]$ and inherit the fixing $F^- = \left\{ \{a, b\} \in F : a, b \in V^- \right\}$.
Construct and solve $\mathcal{LP}(G^-) + F^-$.
if $\mathcal{LP}(G^-) + F^-$ is integral **then**
    Construct H^- from its integral solution.
else
    return (m, x)
end if
Construct a new copy of $\mathcal{LP}(G)$ and impose $x_{ab} = x_{ba} = 1$ for every $\{a, b\} \in F$ and every $ab \in E_{H^-}$.
Solve the modified LP.
if the modified LP is infeasible **then**
    return (m, x)
else
    return the modified LP solution (m, x)
end if

n	H exists		H does not exist	
	LP integral solution	LP fractional solution	LP infeasible	LP fractional solution
3	3	0	1	0
4	6	0	5	0
5	11	0	21	2
6	25	3	117	11
7	62	15	852	115
8	183	124	10271	1768

Table: Outcomes of the vertex deletion linear program using unicliqual vertex as the base for graphs of different orders of n.

n	H exists		H does not exist	
	LP integral solution	LP fractional solution	LP infeasible	LP fractional solution
3	3	0	1	0
4	6	0	5	0
5	11	0	21	2
6	25	3	117	11
7	62	15	852	115
8	194	113	10271	1768

Table: Outcomes of the recursive vertex deletion linear program using unicliqual vertex as the base for graphs of different orders of n.

The linear program only finds a single solution, while there may be many distinct, non-isomorphic solutions. Hence, although fixing a subgraph obtained from the LP may be useful, the particular subgraph selected by the LP may not correspond to the subgraph required to continue the reconstruction of a square root.

n	H exists		H does not exist	
	LP integral solution	LP fractional solution	LP infeasible	LP fractional solution
3	3	0	1	0
4	6	0	5	0
5	11	0	21	2
6	26	2	117	11
7	73	4	852	115
8	274	33	10271	1768

Table: Outcomes of the recursive vertex deletion linear program using unicliqual vertex as the base and enumerating all possible H^- roots for graphs of different orders of n.

Symbolic Sets

Conjecture

Let G and H be graphs such that $G = H^2$. Let $a, b \in V_G$ such that $ab \notin E_G$. For every feasible solution x of $\max \mathcal{LP}(G)$ and every $v \in V_G$ with $v \neq a, b$ we have

$$x_{av} + x_{vb} = 1 \text{ or } x_{av} + x_{vb} = 0.$$

Definition (Consistent Set)

Let $E' \subseteq E_G$ be a maximal subset with $|E'| \geq 3$ such that, for every $e \in E'$, there exists $f \in E'$ such that e and f satisfy the $= 1$ relationship. We call E' a **consistent set** if there exists a partition

$$E' = A \cup B, \quad A \cap B = \varnothing,$$

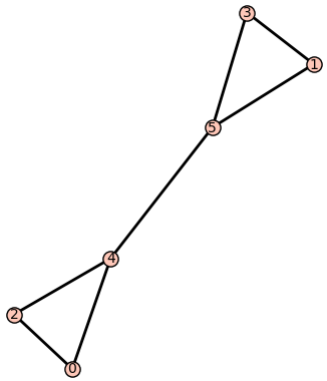
into nonempty sets A and B such that, for all $e, e' \in A$, the edges e and e' do not form a P_3 in G, and for all $f, f' \in B$, the edges f and f' do not form a P_3 in G.

Definition (Inconsistent Set)

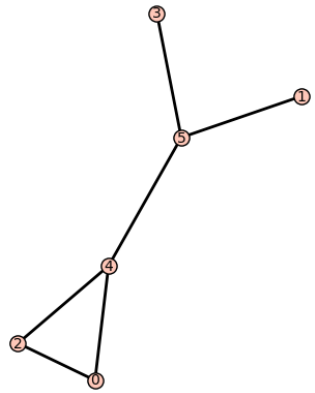
Let $E' \subseteq E_G$ be a maximal subset with $|E'| \geq 3$ such that, for every $e \in E'$, there exists $f \in E'$ such that e and f satisfy the $= 1$ relationship. We call E' an **inconsistent set** if there is no partition

$$E' = A \cup B, \quad A \cap B = \varnothing,$$

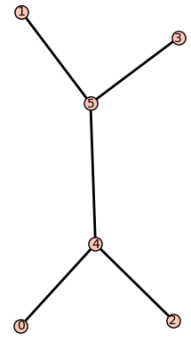
into nonempty sets A and B such that, for all $e, e' \in A$, the edges e and e' do not form a P_3 in G, and, for all $f, f' \in B$, the edges f and f' do not form a P_3 in G.



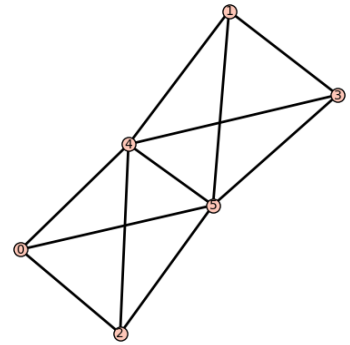
(a) The root graph G_1.



(b) The root graph G_2.



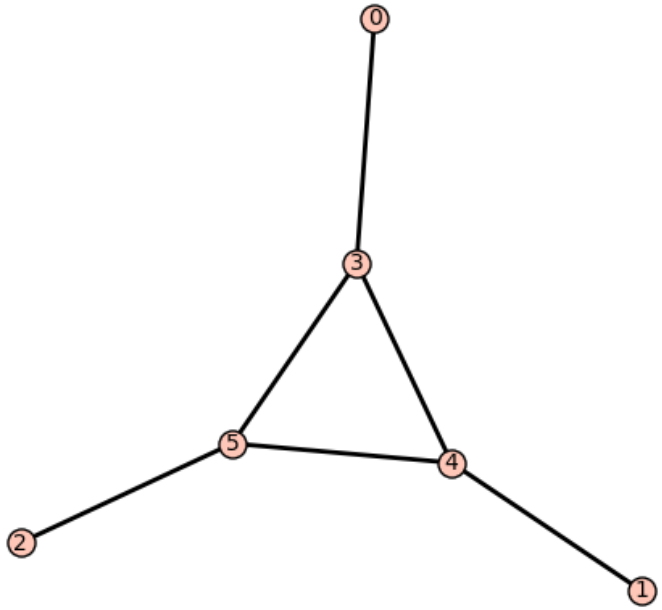
(c) The root graph G_3.



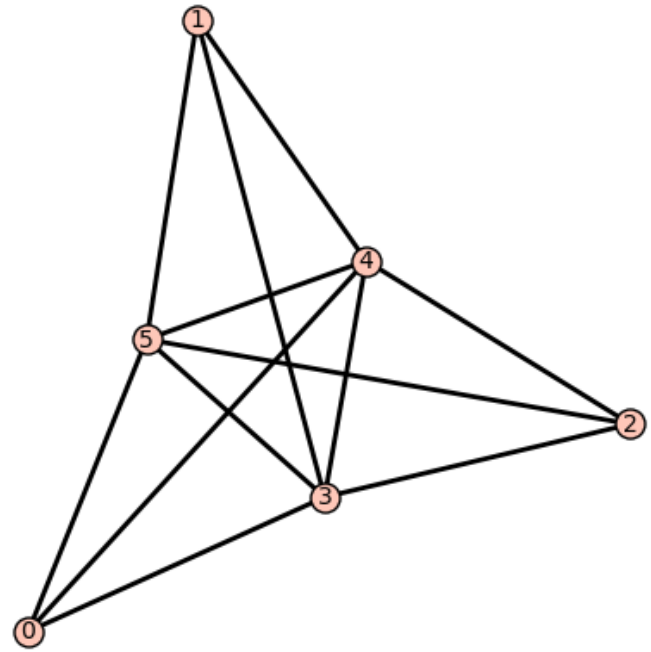
(d) The graph square $H = G_i^2$.

Figure: Three non-isomorphic square roots of the same six-vertex graph square.

$\{(1,4),(4,0),(4,2),(4,3)\}$ and $\{(1,5),(3,5),(5,0),(5,2)\}$ are consistent sets



(a) The root graph G.



(b) The graph square H such that $H^2 = G$.

Figure: A six-vertex graph and its square. G only has one root, although there are different relabellings.

$\{(0,3),(3,1),(3,2)\}$, $\{(0,5),(1,5),(5,2)\}$ and $\{(4,2),(4,1),(4,0)\}$ are inconsistent sets

n	H exists		H does not exist	
	LP integral solution	LP fractional solution	LP infeasible	LP fractional solution
3	3	0	1	0
4	6	0	5	0
5	10	1	20	3
6	26	2	107	21
7	67	10	767	200
8	240	67	9380	2659

Table: Outcomes of the greedy iterative edge enforcement linear program using the max linear program as the base for graphs of different orders of n.

Rounding

1: **Input:** A graph G
2: **Output:** A LP model (m, x)
3: Initialize $F = \emptyset$ and construct $\mathcal{UVLP}(G)$.
4: Solve $\mathcal{UVLP}(G)$.
5: **while** the current LP solution is fractional **do**
6: For each fractional edge ab, compute its rounding score $s_{ab} = |x_{ab} - 0.5|$.
7: Select a fractional edge ab with maximum score.
8: Set

$$\widehat{x}_{ab} = \widehat{x}_{ba} = \begin{cases} 0, & x_{ab} < 0.5, \\ 1, & x_{ab} \geq 0.5. \end{cases}$$

9: Add ab with its assigned value to F.
10: Reconstruct $\mathcal{UVLP}(G)$ with all fixings in F.
11: Solve $\mathcal{UVLP}(G) + F$.
12: **if** $\mathcal{UVLP}(G) + F$ is infeasible **then**
13: **return** the previous feasible solution (m, x).
14: **end if**
15: **if** $\mathcal{UVLP}(G) + F$ is integral **then**
16: **return** the integral solution (m, x).
17: **end if**
18: **end while**
19: **return** (m, x).

n	H exists		H does not exist		Integral $\leftarrow$ Fractional
	LP integral	LP fractional	LP infeasible	LP fractional	
3	3	0	1	0	0
4	6	0	5	0	0
5	11	0	20	3	2
6	27	1	107	21	9
7	73	4	767	200	35
8	245	62	9380	2659	151

Table: Outcomes of the greedy rounding linear program using the max linear program as the base for graphs of different orders of n.

n	H exists		H does not exist		Integral $\leftarrow$ Fractional
	LP integral	LP fractional	LP infeasible	LP fractional	
3	3	0	1	0	0
4	6	0	5	0	0
5	11	0	21	2	1
6	28	0	117	11	6
7	76	1	852	115	25
8	268	39	10271	1768	120

Table: Outcomes of the greedy rounding linear program using the unicliqual vertex linear program as the base for graphs of different orders of n.